Teoler High School

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THAT CAN MAKE ALL THE DIFFERENCE

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CLASS TEST-4

Class : XII-C,D&E **Date : 31-08-2026** **Time : 1½ Hours.** **Max. Marks : 120**

MATHEMATICS

1. Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $\left|\vec{b}\right|^2 = 50$, then $\left|72 - |\vec{b} + \vec{c}|^2\right|$ is equal to ____________.

(1) 66 (2) 88 (3) 106 (4) None of these

2. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors such that $|\vec{a}| = \sqrt{31}, 4\left|\vec{b}\right| = |\vec{c}| = 2$ and $2(\vec{a} \times \vec{b}) = 3(\vec{c} \times \vec{a})$. If the angle between $\vec{b}$ and $\vec{c}$ is $\frac{2\pi}{3}$, then $\left|\frac{\vec{a} \times \vec{c}}{\vec{a} \cdot \vec{b}}\right|^2$ is equal to ______.

(1) 2 (2) 3 (3) 33 (4) None of these

3. Let $\vec{a}$ and $\vec{b}$ be two vectors such that $\left|\vec{a}\right| = 1; \left|\vec{b}\right| = 4,\ \vec{a}.\vec{b} = 2$. If $\vec{c} = (2\vec{a} \times \vec{b}) - 3\vec{b}$ and the angle between $\vec{b}$ and $\vec{c}$ is α, then $192 \sin^2 \alpha$ is equal to______.

(1) 12 (2) 24 (3) 48 (4) None of these

4. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i} - \hat{j} + \hat{k}, \hat{i} + 2\hat{j} + \mu\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$ where $\lambda - \mu = 5,$ are coplanar, then $\sum_{(\lambda,\mu)\in S} 80\left(\lambda^2 + \mu^2\right)$ is equal to :

(1) 2370 (2) 2130 (3) 2290 (4) 2210

5. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = -\hat{i} - 8\hat{j} + 2\hat{k},$ and $\vec{c} = 4\hat{i} + c_2\hat{j} + c_3\hat{k}$ be three vectors such that $\vec{b} \times \vec{a} = \vec{c} \times \vec{a}$. If the angle between the vector $\vec{c}$ and the vector $3\hat{i} + 4\hat{j} + \hat{k}$ is θ, then the greatest integer less than or equal to $\tan^2\theta$ is_________.

(1) 38 (2) 58 (3) 108 (4) None of these

6. The least positive integral value of α, for which the angle between the vectors $\alpha\hat{i}-2\hat{j}+2\hat{k}$ and $\alpha\hat{i}+2\alpha\hat{j}-2\hat{k}$ is acute, is _____.

(1) 55 (2) 15 (3) 5 (4) None of these

7. Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}, \vec{b}=3\hat{i}+4\hat{j}-5\hat{k}$ and a vector $\vec{c}$ be such that $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times\vec{c}=\hat{i}+8\hat{j}+13\hat{k}$. If $\vec{a}.\vec{c}=13$, then $(24-\vec{b}.\vec{c})$ is equal to_____.

(1) 16 (2) 23 (3) 46 (4) None of these

8. Let $\vec{a}=\hat{i}+\hat{j}+\hat{k}, \vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ and $\vec{c}=x\hat{i}+2\hat{j}+3\hat{k}, x\in\mathbb{R}$. If $\vec{d}$ is the unit vector in the direction of $\vec{b}+\vec{c}$ such that $\vec{a}.\vec{d}=1$, then $(\vec{a}\times\vec{b}).\vec{c}$ is equal to:

(1) 11 (2) 3 (3) 9 (4) 6

9. Let $\vec{a}=2\hat{i}+\alpha\hat{j}+\hat{k}, \vec{b}=-\hat{i}+\hat{k}, \vec{c}=\beta\hat{j}-\hat{k},$, where α and β are integers and $\alpha\beta=-6$. Let the values of the ordered pair (α,β), for which the area of the parallelogram of diagonals $\vec{a}+\vec{b}$ and $\vec{b}+\vec{c}$ is $\frac{\sqrt{21}}{2}$, be (α_1,β_1) and (α_2,β_2). Then $\alpha_1^2+\beta_1^2-\alpha_2\beta_2$ is equal to:

(1) 19 (2) 17 (3) 24 (4) 21

10. Let $\vec{a}=9\hat{i}-13\hat{j}+25\hat{k}, \vec{b}=3\hat{i}+7\hat{j}-13\hat{k}$ and $\vec{c}=17\hat{i}-2\hat{j}+\hat{k}$, be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=(\vec{b}+\vec{c})\times\vec{a}$ and $\vec{r}.(\vec{b}-\vec{c})=0$, then $\frac{\left|593\vec{r}+67\vec{a}\right|^2}{(593)^2}$ is equal to______ :

(1) 369 (2) 569 (3) 759 (4) None of these

CHEMISTRY

11. Select complex which have all following characteristics:
- 1 : 1 type electrolyte (number of cation to number of anion ratio)
- Heteroleptic complex
- Does not give AgCl with $AgNO_3$ (solution)

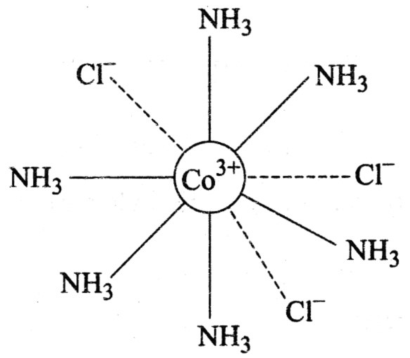
(1) $[Pt(NH_3)_3Cl]Cl$ (2) $K[Pt(NH_3)Cl_5]$
(3) $K_2[PtCl_6]$ (4) $[Pt(NH_3)_2Cl_4]$

12. For which of the following complexes total number of stereoisomers is same irrespective of geometry of complex (C.N. = 4)?
(1) $[M(NH_3)_2(Cl)(Br)]$ (2) $[M(gly)_2]$
(3) $[M(NH_3)(py)(Cl)(Br)]$ (4) $[M(NH_3)_2Cl_2]$

13. $[Co(NH_3)_4(NO_2)_2]Cl$ exhibits:
(1) linkage isomerism, geometrical isomerism and optical isomerism
(2) linkage isomerism, ionization isomerism and optical isomerism
(3) linkage isomerism, ionization isomerism and geometrical isomerism
(4) ionization isomerism, geometrical isomerism and optical isomerism

14. Which of the following is d^2sp^3 hybridisation?
(1) $[NiF_6]^{2-}$ (2) $[IrCl_6]^{3-}$ (3) $[Co(H_2O)_6]^{3+}$ (4) All of these

15. Which of the following complexes (Werner presentation) have minimum electrical conductance in aqueous solution?

(1)


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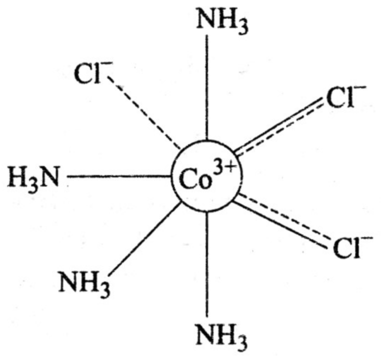
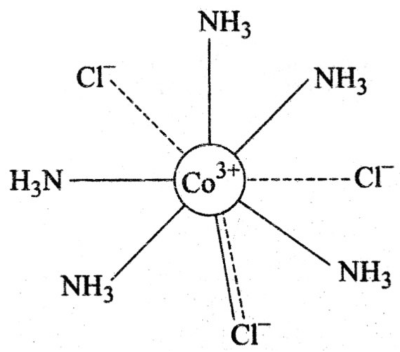
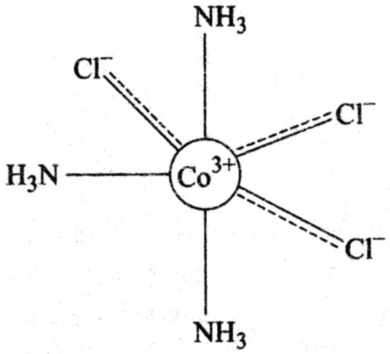


(3)


(4)


16. Which of the following has the highest molar conductivity under similar conditions?
(1) Diamminedichloroplatinum (II) (2) Tetraamminedichlorocobalt (III) Chloride
(3) Potassium hexacyanoferrate (II) (4) Hexaquachromium (III) chloride

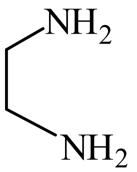
17. Select most stable complex.
(1) $[Co(NH_3)_6]^{3+}$ (2) $[Co(NH_3)_2(en)_2]^{3+}$
(3) $[Co(NH_3)_4(en)]^{3+}$ (4) $[Co(en)_3]^{3+}$

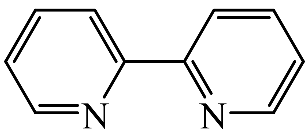
18. After the spliting of d-orbitals in octahedral crystal field, the energy of two eg orbitals will:

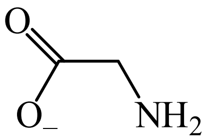
(1) increase by $\left(\frac{3}{5}\right)\Delta_0$ (2) increase by $\left(\frac{2}{5}\right)\Delta_0$

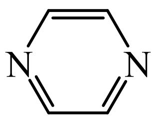
(3) decrease by $\left(\frac{3}{5}\right)\Delta_0$ (4) decrease by $\left(\frac{2}{5}\right)\Delta_0$

19. Which of the following molecule could not act as chelating ligand?

(1) 

(2) 

(3) 

(4) 

20. Select which complex is high spin or spin free octahedral complex.
(1) $[MnCl_4]^{2-}$ (2) $[Fe(H_2O)_6]^{3+}$ (3) $[Co(CN)_6]^{3-}$ (4) $[CrF_6]^{3-}$

PHYSICS

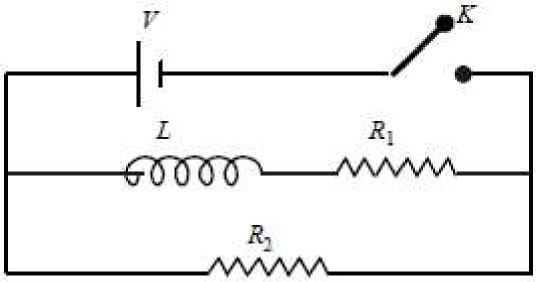
21. A resistor 'R' and 2μF capacitor in series is connected through a switch to 200 V direct supply. Across the capacitor is a neon bulb that lights up at 120 V. Calculate the value of R to make the bulb light up 5 s after the switch has been closed. ($\log_{10} 2.5 = 0.4$)

(1) $1.7\times10^5\Omega$ (2) $2.7\times10^6\Omega$ (3) $3.3\times10^7\Omega$ (4) $1.3\times10^4\Omega$

22. A resistance R and a capacitance C are connected in series to a battery of negligible internal resistance through a key. The key is closed at $t = 0$. If after t sec the voltage across the capacitance was seven times the voltage across R, the value of t is

(1) $3RC\ln2$ (2) $2RC\ln2$ (3) $2RC\ln7$ (4) $3RC\ln7$

23. In the circuit shown below, the key K is closed at $t = 0$. The current through the battery is



(1) $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ $at\ t=0$ and $\frac{V}{R_2}$ $at\ t=\infty$ (2) $\frac{V}{R_2}$ $at\ t=0$ and $\frac{V(R_1+R_2)}{R_1R_2}$ $at\ t=\infty$

(3) $\frac{V}{R_2}$ $at\ t=0$ and $\frac{VR_1R_2}{\sqrt{R_1^2+R_2^2}}$ $at\ t=\infty$ (4) $\frac{V(R_1+R_2)}{R_1R_2}$ $at\ t=0$ and $\frac{V}{R_2}$ $at\ t=\infty$

24. In a LCR circuit capacitance is changed from C to 2 C. For the resonant frequency to remain unchanged, the inductance should be changed from L to

(1) $L/2$ (2) 2 L (3) 4 L (4) $L/4$

25. In an oscillating LC circuit the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is

(1) $\frac{Q}{2}$ (2) $\frac{Q}{\sqrt{3}}$ (3) $\frac{Q}{\sqrt{2}}$ (4) Q

26. A transformer consisting of 300 turns in the primary and 150 turns in the secondary gives output power of 2.2kW. If the current in the secondary coil is 10 A, then the input voltage and current in the primary coil are :

(1) 220 V and 20 A (2) 440 V and 20 A

(3) 440 V and 5 A (4) 220 V and 10 A

27. A coil of inductive reactance 31Ω has a resistance of 8Ω. It is placed in series with a condenser of capacitative reactance 25Ω. The combination is connected to an AC source of 110 V. The power factor of the circuit is

(1) 0.56 (2) 0.64 (3) 0.80 (4) 0.33

28. The primary and secondary coils of a transformer have 50 and 1500 turns respectively. If the magnetic flux ϕ linked with the primary coil is given by $\phi = \phi_0 + 4t$, where ϕ is in weber, t is time in second and ϕ_0 is a constant, the output voltage across the secondary coil is

(1) 90 V (2) 120 V (3) 220 V (4) 30 V

29. In an electrical circuit R, L, C and an AC voltage source are all connected in series. When L is removed from the circuit, the phase difference between the voltage and the current in the circuit is $\pi / 3$. If instead,C is removed from the circuit, the phase difference is again $\pi / 3$. The power factor of the circuit is

(1) 1/2 (2) $1/\sqrt{2}$ (3) 1 (4) $\sqrt{3}/2$

30. A coil of self-inductance L is connected in series with a bulb B and an AC source. Brightness of the bulb decreases when

(1) frequency of the AC source is decreased

(2) number of turns in the coil is reduced

(3) a capacitance of reactance $X_C = X_L$ is included in the same circuit

(4) an iron rod is inserted in the coil

Answer Key

Class : XII-C,D&E **CLASS TEST- 4** **Date : 31-08-2026**

(ON LINE TEST)

MATHEMATICS									
1	2	3	4	5	6	7	8	9	10
1	**2**	**3**	**3**	**1**	**3**	**3**	**1**	**1**	**2**
CHEMISTRY									
11	12	13	14	15	16	17	18	19	20
2	**2**	**3**	**4**	**4**	**3**	**4**	**1**	**4**	**2**
PHYSICS									
21	22	23	24	25	26	27	28	29	30
2	**1**	**2**	**1**	**3**	**3**	**3**	**2**	**3**	**4**